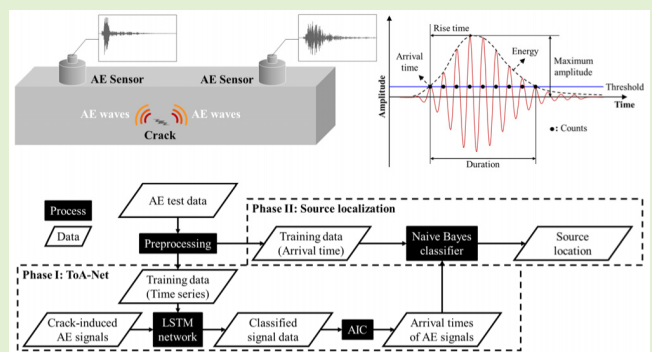


A Hybrid Approach of Long Short-Term Memory and Machine Learning With Acoustic Emission Sensors for Structural Damage Localization

Yunwoo Lee^{ID}, Jae Hyuk Lee, Jin-Seop Kim^{ID}, and Hyungchul Yoon^{ID}

Abstract—Various sensors are used for structural health monitoring (SHM). An acoustic emission (AE) sensor detects an elastic wave propagating in the medium, so it can detect the possibility of defects occurring inside the structure. Using multiple sensors enables the estimation of the signal source through the differences in signals measured by each sensor. Among the information for signal analysis, the time difference of arrival is the most commonly used factor for estimating the location of the source. However, it is difficult to accurately determine the arrival time because the measured signal always contains ambient noise. Even though the arrival times of signals are determined, there is the following task to identify the source location, which is also complicated because the signal does not propagate with a constant velocity throughout the medium. To solve this problem, this study adopts a hybrid approach that applies artificial intelligence techniques step by step. In the first phase, the time-series data are classified as signal and nonsignal by the long short-term memory (LSTM) network. The second phase is to identify the source location based on the naive Bayes classifier using the distribution of the arrival times of signals extracted from multiple sensors. Since this approach reduces complex computations in signal processing while minimizing the masking of physical meaning by black-box AI technology, it allows for versatile applications depending on the objectives. The proposed method was validated through an experimental test, and the results showed that the method had reliable performance.

Index Terms—Acoustic emission (AE), damage detection, long short-term memory (LSTM), naive Bayes classifier, structural health monitoring (SHM).



I. INTRODUCTION

FOR major infrastructures, the condition of the structure is evaluated regularly to ensure its safety. In general, the evaluation process includes manual inspection to check the performance of the structure at regular intervals and sensor-based monitoring that checks for abnormal behavior.

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Structural health monitoring (SHM) is a method for continuously monitoring the integrity of structures, and various sensors can be used for the purpose [1], [2], [3], [4], [5]. The measured structural behaviors should be able to represent the state of the structure directly or indirectly, and the condition of the structure can be estimated through various methods according to the characteristics of the measured data [6], [7], [8], [9], [10]. In modern SHM systems, the real-time and accurate detection of structural damages is crucial for ensuring public safety. However, traditional monitoring techniques often struggle to effectively identify internal defects in complex structures or under high-noise environments, necessitating new, more reliable methods.

When a crack-like defect occurs inside a structure, it is accompanied by energy release owing to plastic deformation (or destruction) of the solid. Such a signal propagating in the form of elastic waves from a localized source is called an acoustic emission (AE) signal, and the sensor that can detect this signal at the surface of the structure is called an AE sensor. While ultrasonic technology uses the reflecting characteristics

of the signal to detect the existing internal defects, AE technology analyzes the propagating characteristics of the signal generated by the defect. This allows real-time monitoring of defects occurring inside the structure. Based on its strengths, AE technology has been used widely not only in SHM [11], [12], [13] but also in various fields, such as machine health monitoring, fluid leak detection [14], [15], [16], and even in electronic components, such as metal–oxide–semiconductor field-effect transistors (MOSFETs), where stress waves are analyzed to assess the working state of the device [17], [18], [19].

By analyzing the AE signals, the location or type of defect can be estimated. The most general method for analyzing AE signals is to estimate the location of the defect using the time difference between the signals arriving at multiple sensors. However, estimating the location of the defect is not a simple problem because various influencing factors are acting in a complex way, such as the material properties of the structure and the propagation characteristics of elastic waves. To estimate the location of defects using multiple AE sensors, it is essential to accurately determine the arrival time of the signal, because the differences in arrival times can be regarded as the differences in the transferred distances. A general approach for determining the arrival time of a signal is to use a threshold. This approach assumes that the valid signal is sufficiently large to be distinguished from background noise. The arrival time can then be considered as the time when the signal first exceeds the threshold. However, this approach contains significant error when the noise level is high [20].

To calculate the arrival time more accurately, the Akaike information criterion (AIC) was introduced [21], [22]. It divides time-series data into locally stationary segments using an autoregressive (AR) model. This approach separates the signal and noise to determine the arrival time of the signal, which is called the AR-AIC picker [23], [24]. Maeda [25] proposed a function to calculate the AIC directly from the time series without using AR coefficients. Although the AIC performs well when the signal is clearly distinguished from the noise, it contains a large error when the signal-to-noise ratio is low [26]. There are various studies that have been conducted to overcome the limitations on the application of AIC [27], [28], [29]. However, the methods are not generalized enough to be applied to other data because they use the characteristics of the data used in that particular study.

Once the arrival times of the signals are extracted from multiple sensors, the time information can be used to estimate the source location. There are two types of approaches to source localization based on the arrival time of signals: the zonal location and point location methods [30], [31]. The zonal location method identifies the source location as one of the predefined zones, whereas the point location method estimates the x , y , and z coordinates of the source. Although the point location method can provide more generalized solutions, it is difficult to find accurate source locations in the case of nonhomogeneous materials or complicated structures. The methodology for zonal location originated from studies in materials science [32] and has the advantage that it can

be simply applied to estimate the approximate location of the source. It determines the zone from which the signal is supposed to have been generated by subdividing the zone into subzones according to the arrival sequence (time) of the signal and the position of sensors [33]. However, because the method works based on the priority of the measured information, it is difficult to estimate the location accurately when the information is not in order owing to the difference in the signal propagation path.

Recently, researchers have been making efforts to address the challenges in conventional methods of structural damage localization by adopting AI technologies for AE signal analysis. From a methodological perspective, artificial neural networks (ANNs) [34], [35], which have a simple network structure, or convolutional neural networks (CNNs) [36], [37], which can be applied by enhancing the features of the signal through time–frequency analysis, have been mainly used. The AI-based approach involves constructing a black-box model that is determined through iterative learning, and it can provide alternative solutions to complicated problems based on high-quality data. However, as the problem becomes more complicated, it becomes more difficult to interpret the solution derivation process and apply it to other untrained conditions. The methodology only has input data and output results, and the physical meaning of the intermediate process is hidden as a black box.

To address these limitations, this study proposes a novel AE signal analysis method that enhances interpretability while maintaining the accuracy and efficiency of AI techniques. Unlike conventional AI approaches that operate as black-box models for the entire process, the proposed method applies AI models to each minimal step of analysis, ensuring the preservation of physical meaning at each stage. This method not only maintains the integrity of subresults but also improves the overall convenience of the process. It leverages long short-term memory (LSTM) networks, which retain memory overextended sequential analyses, and a naive Bayes classifier, which provides probability-based solutions through feature distribution. The proposed method comprises two key steps: 1) automatic extraction of AE signal arrival times and 2) source location estimation based on arrival time differences. To detect AE signal arrival times, an LSTM-based ToA-Net was developed, classifying AE data into signals and nonsignals, with the arrival time determined using AIC from the classified data. A naive Bayes classifier then estimates the source location using arrival times measured by multiple sensors. As each step sequentially processes data and identifies the most probable signal location, the method maintains interpretability and preserves the physical meaning throughout the signal processing.

This article is structured as follows. Section II describes the basic theoretical background of the AE signal analysis method. Section III presents the methodology proposed in this study, which consists of two parts: the arrival time extraction of the signal and the source localization. In Section IV, the feasibility of the proposed method is evaluated through a validation test, and Section V covers a brief conclusion and future work for the study.

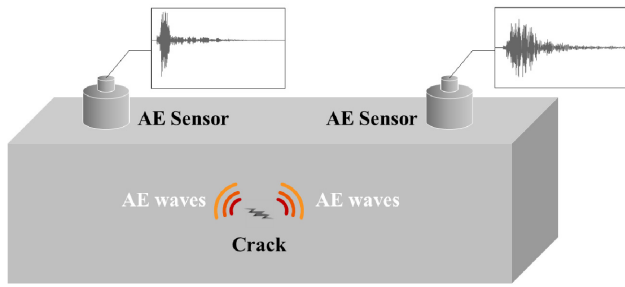


Fig. 1. Basic concept of AE technique.

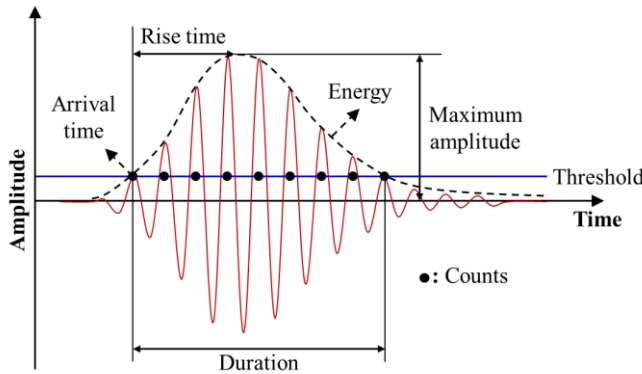


Fig. 2. Typical parameters of AE signals.

II. THEORETICAL BACKGROUND

When a structural defect, such as a crack, is initiated inside a structure, it is accompanied by the emission of elastic waves due to the plastic deformation of the solid (Fig. 1). The dominant frequency range of the generated AE signal is several kilohertz to several megahertz [38], [39], [40] and can be detected and converted into an electric signal by the AE sensor. The extracted data can be analyzed in various ways to obtain information related to defects. Fig. 2 shows the typical AE signal parameters that can be used for the analysis. Parameters, such as maximum amplitude, rise time, hit counts, and envelope, can be used for signal analysis to determine the types and locations of the defects [41], [42], [43].

When using multiple sensors, the source-related information can be estimated through the difference in data from each sensor because the extracted characteristics are different even for the signal generated from one source. The most general characteristic of this method is the arrival time of a signal. Assuming that the medium is homogeneous, the signal propagates at the same speed, and the source location can be estimated from the difference in the distance from each sensor. In a simple manner, the arrival time of the signal can be determined based on a certain threshold for the amplitude. However, it is not robust and can lead to abnormal values when large amounts of noise or outliers are included. Methods other than threshold-based ones are mainly based on two types of approaches. One is the short-term average/long-term average (STA/LTA) method, which uses a characteristic function that considers changes in amplitude and frequency. The other is an AIC, which uses an AR model to determine changes in the signal. This study is based on the latter.

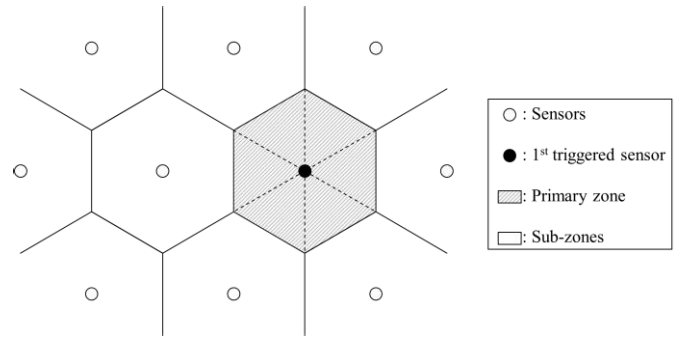


Fig. 3. Zonal location method.

When using the AIC, an AR model is used to divide the signal into two locally stationary segments. The AIC for the signal of length N is calculated as a function of merging point k in (1), where M is the order of the model, C is a constant, and $\sigma_{1,\max}^2$ and $\sigma_{2,\max}^2$ are the variances of the prediction errors for each range

$$\text{AIC}(k) = (k - M) \cdot \log(\sigma_{1,\max}^2) + (N - M - k) \cdot \log(\sigma_{2,\max}^2) + C. \quad (1)$$

The order of the models was determined through trial and error, and the coefficients of the AR model were calculated for both segments. Because AIC indicates the fitness of the AR models, the point at which AIC is minimized in the fixed-order AR process can represent the optimal separation point of the signal. The separated segments can be regarded as signals and nonsignals, and the time point at which the signal is separated can represent the arrival time of the signals. However, the calculation of the AR coefficients and the order in AR-based AIC has a high degree of computational complexity. In contrast, Maeda (1985) suggested a method to calculate the AIC directly without using AR coefficients, as shown in the following equation:

$$\text{AIC}(k) = k \cdot \log(\text{var}(x[1, k])) + (N - k - 1) \cdot \log(\text{var}(x[k + 1, N])) \quad (2)$$

where $\text{var}(x[1, k])$ and $\text{var}(x[k + 1, N])$ are the variances of the time series for the ranges up to and after time k , respectively. AIC is a method for finding the point at which the before and after characteristics of the data can be differentiated.

The arrival times of the signal extracted using multiple sensors can be used to determine the source location through zonal or point location methods. The zonal location method is used to identify whether the signal is from a predefined zone, based on the first triggered sensor. In general, the zone is divided according to the distribution of the sensors, as shown in Fig. 3, and then, the primary zone is determined based on the position of the sensor that was first triggered. It then subdivides the zone into subzones using the information from sensors that are triggered later to determine the zone from which the signal originated. The method can be applied simply and shows more rational performance as the network is denser and can be the only alternative, particularly in cases where it is difficult to locate the source, such as in complex structures.

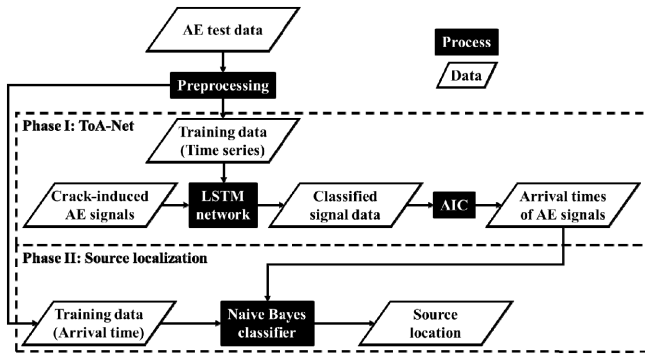


Fig. 4. Schematic description of the proposed method.

The point location method can be applied based on (3), which is a relation between the positions of the sensor and source

$$\sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2} = v(t_i - t) \quad (3)$$

where x , y , and z are the coordinates of the source, and x_i , y_i , and z_i are the coordinates of the i th sensor. v is the propagation velocity, and t and t_i represent the origin time of the event and the arrival time at the i th sensor, respectively.

Basically, the point location method requires more than a certain number of sensors. The equation can be solved using iterative or noniterative methods. Although the noniterative method has advantages of being fast and easy to apply since it does not require a numerical approach, there is a limitation in that it lacks flexibility. On the other hand, the iterative method may be accompanied by computational issues, such as the convergence of solutions, but it can be used flexibly for complex problems compared to the noniterative method.

III. METHODOLOGY

A. Overview

Fig. 4 shows the schematic of the proposed method. The AE test data are preprocessed according to the purpose and are applied for training an LSTM network in the ToA-Net and a naive Bayes classifier. The input of the LSTM network is time-series AE data, and the output is the class (signal and nonsignal) of the data at each time step. By applying AIC to the classified signal data, the arrival times of the signal are determined in multiple sensors, and the naive Bayes classifier identifies the location of the source using a statistical method based on the pretrained arrival time distribution.

B. ToA-Net

The primary step for identifying the main information regarding the distance between the source and sensor is to extract the arrival time of the signals. Methods, such as STA/LTA and AIC, can be used to determine the arrival times of signals. However, the above methods have limitations in that they are difficult to generalize for application because the time-series data may have different characteristics depending on the signals. This study used the LSTM network to identify the arrival times of the AE signals. Because the LSTM network can consider the sequential relation of data, it can be used to classify the ranges of signals and nonsignals (i.e., noise) from

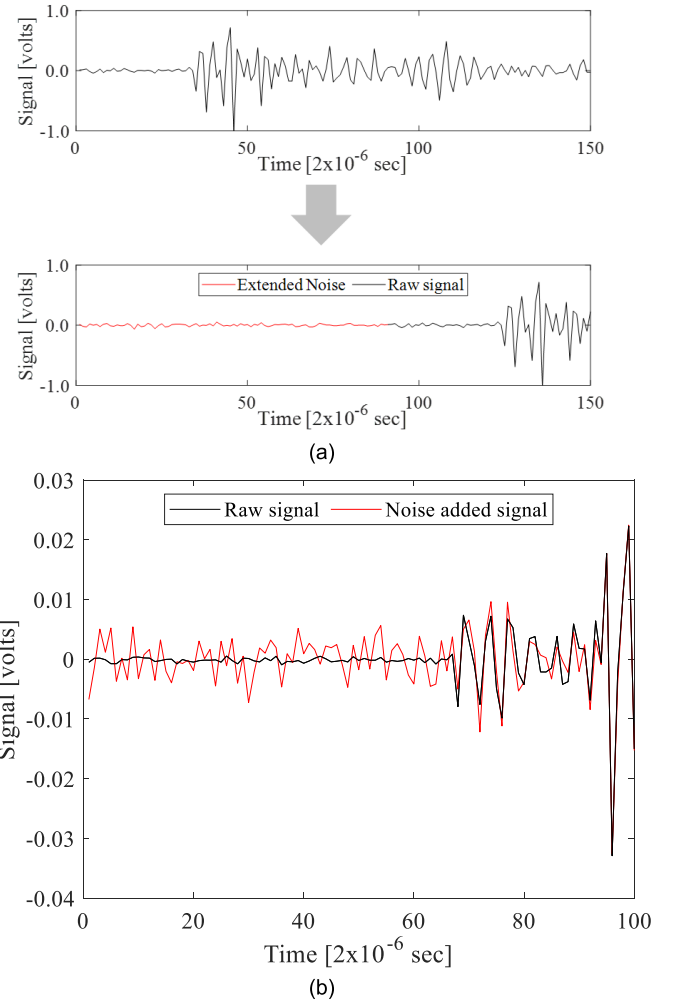


Fig. 5. Preprocessing. (a) Extending pretrigger length. (b) Extending pretrigger length.

time-series data by analyzing the sequential characteristics of the data. Once the measured AE data are classified as signal and noise, the time at which the classes are separated can be regarded as the signal arrival time.

1) *Preprocessing Data*: The input data are used in the LSTM network to learn the data as signals and nonsignals. For the variety of learning conditions and the robustness of the network, two types of preprocessing were applied on the data. One is to extend the pretrigger range, that is, the range before the signal arrives. For this purpose, an arbitrary length of white Gaussian noise scaled to the noise levels of the pretrigger ranges of each data was added prior to the data, as shown in Fig. 5(a).

The other preprocessing step was applied to increase the overall noise level of the signal. In this process, white Gaussian noise was added to the entire data range, and its magnitude was determined based on the standard deviation of the original signal. In this study, the training data were preprocessed by applying noise of 5% and 10%. Fig. 5(b) shows an example of the trigger range for data with a noise of 10%.

2) *Network Configuration*: This study used the LSTM network, which is an extension of the RNN, to analyze the AE signals. Contrary to the conventional feed-forward network,

an RNN can consider the sequential characteristics of data by applying the output due to one input as the input to the next sequence. Fig. 6(a) shows the simplified data processing in the RNN model. The values derived in the hidden layer of the input data are not only transmitted to the output layer but also affect the determination of the value in the hidden layer of the next sequence. Let $x = (x_1, x_2, \dots, x_T)$ be the sequential inputs. Then, its value in the hidden layer, that is, the hidden state vector, is calculated as in the following equation:

$$h_t = \sigma(W_x x_t + W_h h_{t-1} + b_t) \quad (4)$$

where σ denotes an activation function (e.g., sigmoid, hyperbolic tangent, or rectified linear unit function), W_x and W_h represent the weight matrices for x and h , respectively, and b_t is the constant bias vector. The network updates the weights and biases at each node through backpropagation, using the derivative of the activation function to derive the intended output. As the sequence length increased, the parameters were not updated because the gradient gradually converged to zero as the calculation was repeated. This is called the “vanishing gradient,” and LSTM was developed to overcome this limitation [44]. As shown in Fig. 6(b), the LSTM cell consists of three gates, namely, the forget, input, and output gates. The forget gate determines how much information from the past is forgotten or transmitted using (5). In this equation, W_f and b_f represent the weight and bias for the forget gate, respectively. The hidden state vector of the previous time step (h_{t-1}) is combined with the current input (x_t), and the output (f_t) is derived through the sigmoid function (σ) of (6)

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}. \quad (6)$$

The input gate uses the sigmoid and hyperbolic tangent functions, as in (7) and (8), to determine the information to be stored among the current inputs. W_i and b_i and W_c and b_c represent the weight and bias for the input gate and cell state, respectively. The outputs of the forget and input gates are combined with the information from the previous cell (c_{t-1}) in the cell state to determine the information of the current cell, as shown in (9)

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (7)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (8)$$

$$c_t = f_t * c_{t-1} + i_t * \tilde{c}_t. \quad (9)$$

The output gate combines the current information with past information and transmits it to the hidden layer through (10) and (11), where W_o and b_o are the weight and bias at the output gate. The sigmoid of the previous hidden state vector and the current input is multiplied by the hyperbolic tangent of the current cell state vector to derive the current hidden state vector

$$\sigma_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (10)$$

$$h_t = \sigma_t * \tanh(c_t). \quad (11)$$

The values from the LSTM layer are transmitted to the fully connected layer and then classified into predefined classes

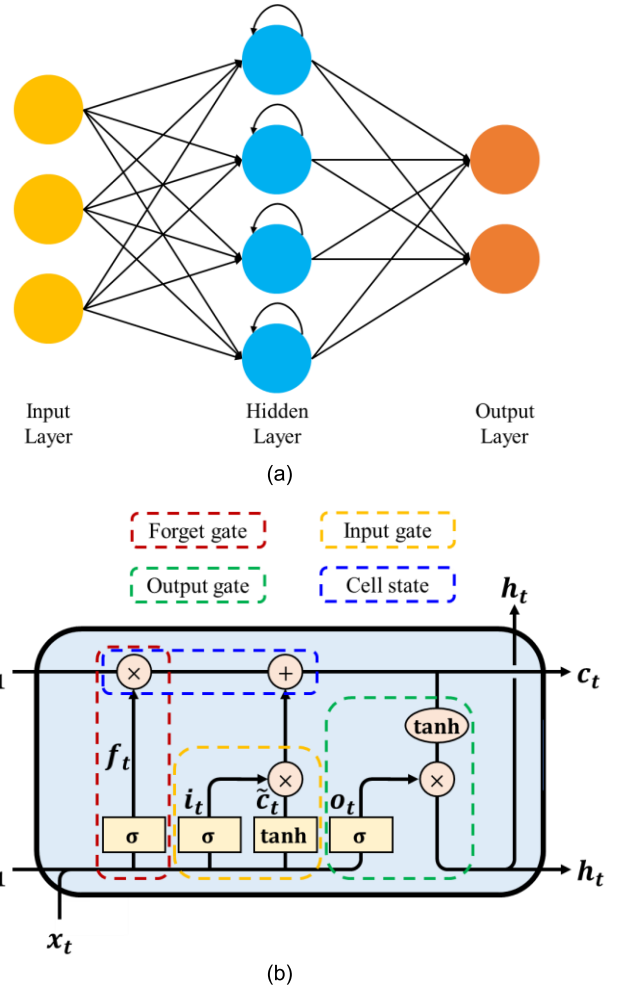


Fig. 6. Schematic description of (a) RNN and (b) LSTM.

using the softmax function of (12), where z_i is the element of the input vector and K is the number of classes. In this study, two classes of “nonsignal” and “signal” were predefined for the data before and after the arrival of the signal, respectively

$$\text{softmax}(z)_i = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)}. \quad (12)$$

3) Training Network: The data labeled as “signal” and “nonsignal” classes based on the arrival time were applied to the LSTM network. A preliminary analysis was performed to check the influence of training conditions on the network. At low learning rates, the learning proceeded stably, but it was difficult to arrive at the best solution, whereas, at high learning rates, it was unstable but easily led to a better solution. For a batch size, the learning proceeded more stably when a larger batch size is applied than when a smaller batch size is applied. Based on these results, in this study, the network was trained with a learning rate of 0.001 and a batch size of 128. The maximum number of epochs was set to 500, and the early stopping option to prevent overfitting was set to 100.

Among the preprocessed data, 80% was used as an input for training, 10% was used for validation of the training process, and 10% was used for testing the trained network. Network performance was assessed by the number of misclassified data,

TABLE I
CONSIDERED PARAMETERS IN THE PARAMETRIC STUDY FOR
THE LSTM NETWORK

Number of hidden layers	Number of hidden units
1	100
2	200
3	300
4	400
5	500

and a parametric study was performed for the hyperparameters (the number of hidden layers and hidden units) of the network. Table I lists the parameters considered. The number of hidden layers and hidden units were considered from 1 to 5 and 100 to 500, respectively. The performance, which is measured by the average number of misclassified data points for the network for the given parameters, is shown in Fig. 7(a). The results show that the networks with the considered parameters exhibit an average of approximately 1.5 misclassified data points over time-series data of length around 1000, verifying their effectiveness in classifying data into signal and nonsignal segments. It was found that the difference in the average number of misclassified data points between the simplest and the most complex network structures is only 0.07, and the difference between the highest and lowest performing networks is only 0.26.

The LSTM network used to classify AE data into signal and nonsignal assigns a class to each data point, which means that it can generate discrete outcomes rather than a binary distinction between signal and nonsignal. This indicates that the network became more sensitive to the sequential pattern of data for classification. Therefore, such characteristics of the trained network were analyzed by examining the distance between the first data point classified as “signal” and the last data point classified as “nonsignal.” The results of this analysis for the parameter study are illustrated in Fig. 7(b). The results indicate that the network produces discontinuous outcomes of about 2.5-time steps on average when classifying AE data into signal and nonsignal. This discontinuity can interfere with the accurate identification of signal arrival times, highlighting the need for a postprocessing step to refine these estimates.

4) *Postprocessing Using Modified AIC*: In previous studies on the determination of arrival time of signals, the AIC is one of the most popular methods. It divides the data into segments of noise and signals. Although it performs well when the data are clearly distinct, such as when only two types of signals are included, it is generally difficult to find the intended point because there are various influencing factors, as shown in Fig. 8.

In this study, the time-series data of AE signals were classified into signal and nonsignal through the LSTM network. Because LSTM classifies the class of signals for every time step based on the sequential information of the data, the result may not be completely divided into two parts: a signal and a nonsignal section. As shown in Fig. 9, discontinuous outputs can be observed in the classified results. In this regard, this study assigns a binary value to the classified output and determines the arrival time of the signals by using it

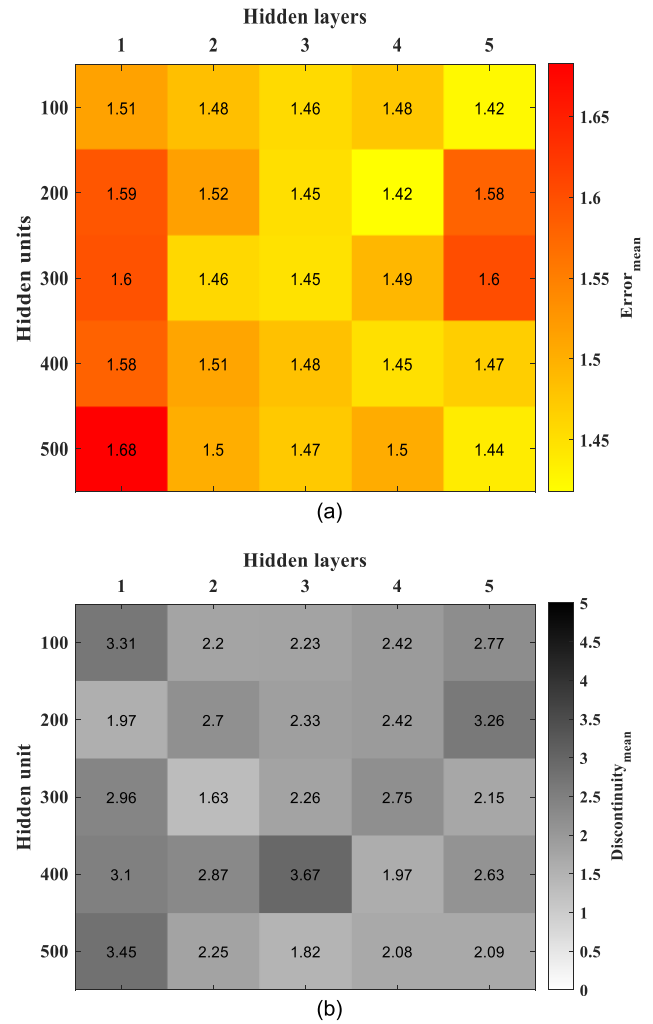


Fig. 7. Results of the parametric study. (a) Mean of misclassified data. (b) Discontinuity in the classified results.

for postprocessing based on the AIC. However, because the variance of data assigned a binary value can be zero, the modified AIC as in (13), in which log is excluded from (2), is used in this stage

$$\text{AIC}(k) = k \cdot \text{var}(x_c[1, k]) + (N - k - 1) \cdot \text{var}(x_c[k + 1, N]) \quad (13)$$

where x_c represents the classified data, assigned a binary value. Fig. 9 shows an example of applying a modified AIC to the binary value-assigned classified output, where the point at which the AIC is the minimum represents the arrival time of the signals.

C. Source Localization

When a defect occurs inside a structure, identifying its location is an important task for follow-up. The AE sensor detects an acoustic signal derived from the occurrence of a defect, making it possible to estimate the location of the defect. Assuming that the distance and time for signal propagation are proportional, the location of the source can be estimated based on the time difference of the signals arriving at multiple

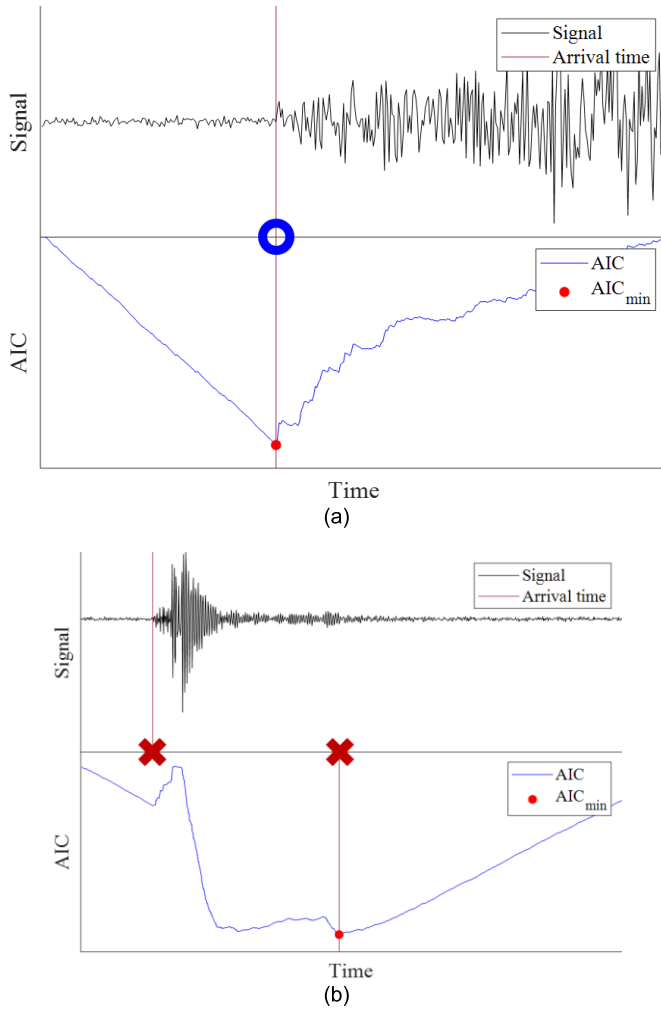


Fig. 8. Application of AIC. (a) Ideal case. (b) General case.

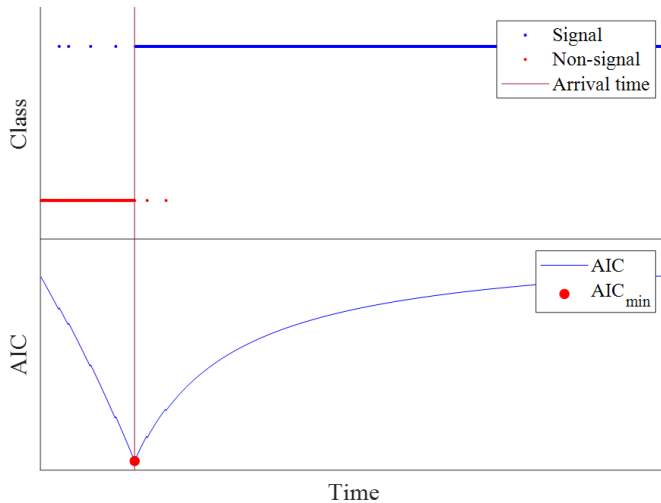


Fig. 9. Application modified AIC to the classified output.

sensors. However, there is an inevitable error because the signal does not propagate with a constant velocity throughout the medium, and a higher level of knowledge about the characteristics of signal propagation and materials is required for accurate estimation.

This study used the naive Bayes classifier, a probability-based machine learning algorithm, to determine the location of the AE source. This method is based on Bayes' theorem, which represents the probabilistic relationship between variables in a dependent relation. When the prior and likelihood probabilities are $P(A)$ and $P(B|A)$, respectively, the posterior probability $P(A|B)$ is expressed as in the following equation:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}. \quad (14)$$

Assuming that there are some available features ($x_1, x_2, \dots, x_n$) for the instance to be known, the probability of the instance can be expressed as in (15), where n is the number of statistically independent features and $P(C_k)$ is the probability of k possible classes

$$P(C|x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i|C_k) \frac{P(C_k)}{P(x_1, x_2, \dots, x_n)} \quad (15)$$

where $P(C_k)$, $P(x_i|C_k)$, and $P(x_1, x_2, \dots, x_n)$ represent the prior, likelihood, and evidence probabilities, respectively. $P(C_k)$ and $P(x_i|C_k)$ can be obtained from the training data, and $P(x_1, x_2, \dots, x_n)$ are applied as constants in the machine-learning algorithm. As a result, the naive Bayes classifier finds C_k corresponding to class k , which has the maximum probability among the classes, as shown in the following equation:

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} P(C_k) \prod_{i=1}^n P(x_i|C_k). \quad (16)$$

The input data for the naive Bayes classifier were modeled as a probability density function (pdf) based on kernel density estimation (KDE), as shown in (17). The kernel function was used as a Gaussian distribution, as in (18), and the bandwidth parameter h was determined based on (19), which minimizes the mean integrated squared error [45]

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n K_n(x - x_i) = \frac{1}{nh} K\left(\frac{x - x_i}{h}\right) \quad (17)$$

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} \quad (18)$$

$$h = \left(\frac{4\sigma^5}{3n}\right)^{\frac{1}{5}} \approx 1.06\sigma n^{-1/5}. \quad (19)$$

In this study, the naive Bayes classifier uses the possible AE source location divided into several zones as a class and the arrival times of signals from multiple sensors as training data. Because the exact time for a signal to propagate from the source to the sensor is unknown, the time difference of arrival was used as the input based on the arrival time of the first triggered sensor as 0. The naive Bayes classifier probabilistically derives the most suitable zone as the source location through the distribution of the arrival times of the sensors.

IV. VALIDATION TEST

A. Validation Test Setup

The test for the AE signal analysis was performed on a concrete panel specimen, as shown in Fig. 10. The width,

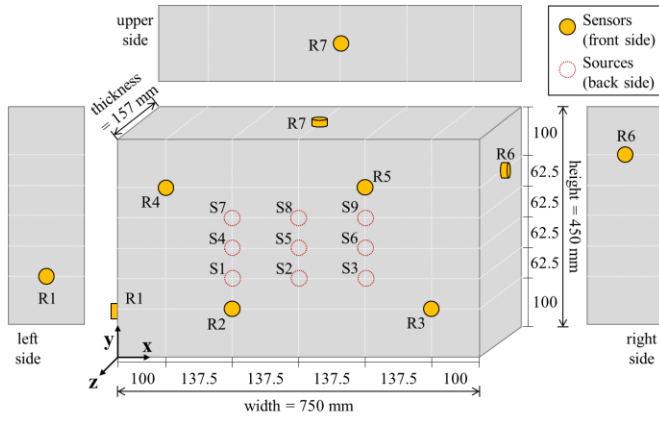


Fig. 10. Experimental test setup.

TABLE II
COORDINATES OF SENSORS IN AE TEST

	x [mm]	y [mm]	z [mm]
Sensor #1	0	100	-78.5
Sensor #2	237.5	100	0
Sensor #3	650	100	0
Sensor #4	100	350	0
Sensor #5	512.5	350	0
Sensor #6	750	350	-78.5
Sensor #7	375	450	-78.5

TABLE III
COORDINATES OF SOURCES IN AE TEST

	x [mm]	y [mm]	z [mm]
Source #1	237.5	162.5	-157
Source #2	375	162.5	-157
Source #3	512.5	162.5	-157
Source #4	237.5	225	-157
Source #5	375	225	-157
Source #6	512.5	225	-157
Source #7	237.5	287.5	-157
Source #8	375	287.5	-157
Source #9	512.5	287.5	-157

height, and thickness of the specimens were 650, 450, and 157 mm, respectively. AE sensors were attached to the front, side, and top surfaces of the specimen and the source location was set to be distributed on the back side. Table II lists the coordinates of the AE sensors, and Table III lists the coordinates of the source locations. Acoustic signals were generated by a pencil lead break, and the test was performed 50 times for each source location. From the experimental test, 3150 ($9 \times 7 \times 50$) data were obtained, and a total of 9450 (3150×3) data were used for the study by considering the cases, where 5% and 10% of noise were added together. The sampling rate of the AE sensor used was 500 kHz, and the signal was measured for approximately 2 ms (i.e., the length of the single data was approximately 1000).

The data were obtained based on the 3σ threshold value for the noise range and saved to have a pretriggered length of approximately 60 μs . For the training data, the arrival time of the signal was determined by applying AIC to the range,

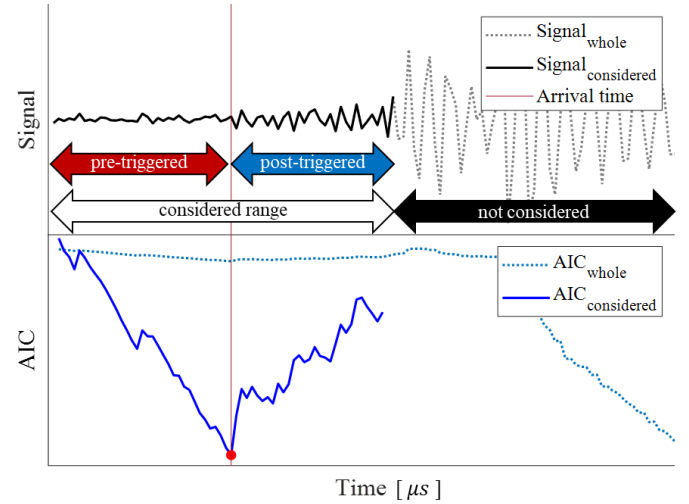


Fig. 11. Determination of the arrival time for training data.

in which pretriggered and posttriggered signals of the same length were considered. In cases where the approximate arrival time range of the signal was known, the AIC could derive a relatively accurate solution, but not when considering the whole signal, as shown in Fig. 11. For some data whose arrival time was not well distinguished by the AIC owing to a high noise level, the arrival time was picked manually. The training data were applied to the LSTM network as signal or nonsignal based on the determined arrival time.

B. Arrival Time Extraction

Based on the parametric study, the number of hidden layers and hidden units for the ToA-Net were determined to be 4 and 200, respectively, which showed the best performance among the analyzed parameters. The data classified by the LSTM network are applied to (13), and the point with the minimum AIC value is determined as the arrival time of the AE signals. Fig. 12 shows the result of the extracted arrival times for the AE signals using ToA-Net. The result was compared with conventional statistical threshold methods. When applying the 3σ threshold, many false detections occurred, where the extracted arrival time was earlier than the actual signal arrival time, due to significant noise relative to the standard deviation of the signals. Applying the 6σ threshold can reduce the early detection errors in signal arrival time that occurred with the 3σ threshold. However, the increased threshold made it difficult to detect the signal arrival until the signal amplitude increased beyond a certain level after the actual signal had already arrived. The proposed method achieved a root mean square error (RMSE) of approximately 5.7548 μs for the actual signal arrival time and an R^2 value of approximately 0.9905 with respect to $y = x$, demonstrating superior performance compared to the conventional methods.

Fig. 13 shows the error distributions for the locations of the sensors and sources. Regarding the location of the sensors, the data measured from sensor 5 showed the most accurate results, with an RMSE of approximately 2.4221 μs , and the error for sensor 4 was the largest, with an RMSE of approximately 12.4936 μs . In terms of source location, the

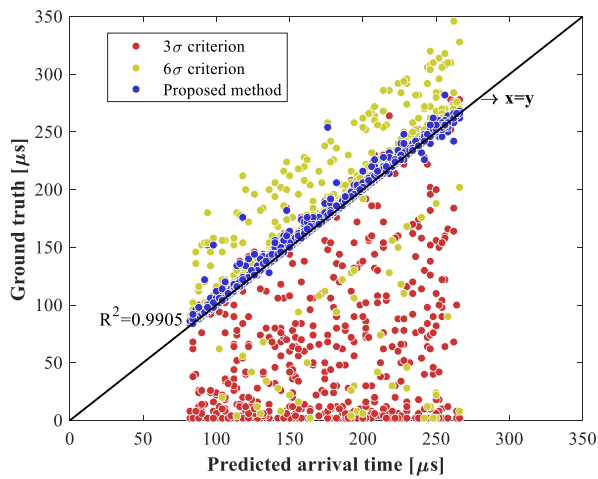


Fig. 12. Arrival time extraction results of ToA-Net.

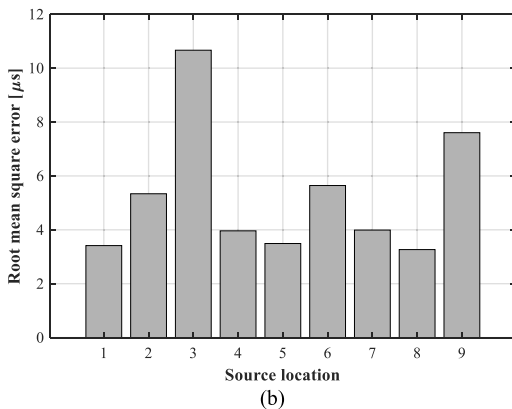
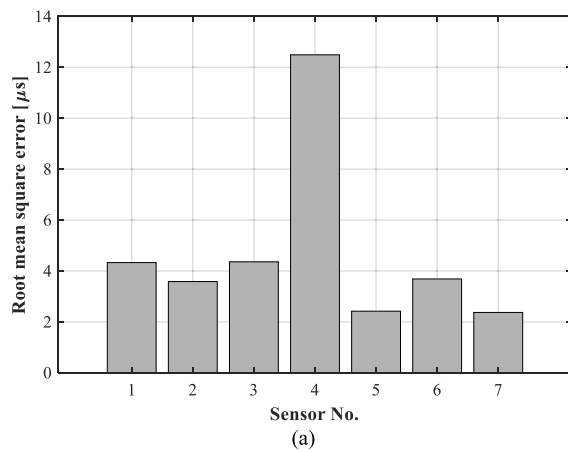


Fig. 13. Error of ToA-Net for the location of (a) sensors and (b) sources.

AE signal generated from source 8 showed the most accurate result, with an RMSE of approximately $3.2660 \mu\text{s}$, whereas the signal from source 3 showed the largest error, with an RMSE of approximately $10.6655 \mu\text{s}$. Although the signals for all cases were used in equal proportions for training the LSTM network, the results showed deviations depending on the sensor and source locations. The factors that cause such differences could be the nonhomogeneity of materials on the propagation path of AE signals, the installation state of sensors, and differences in the data collection process.

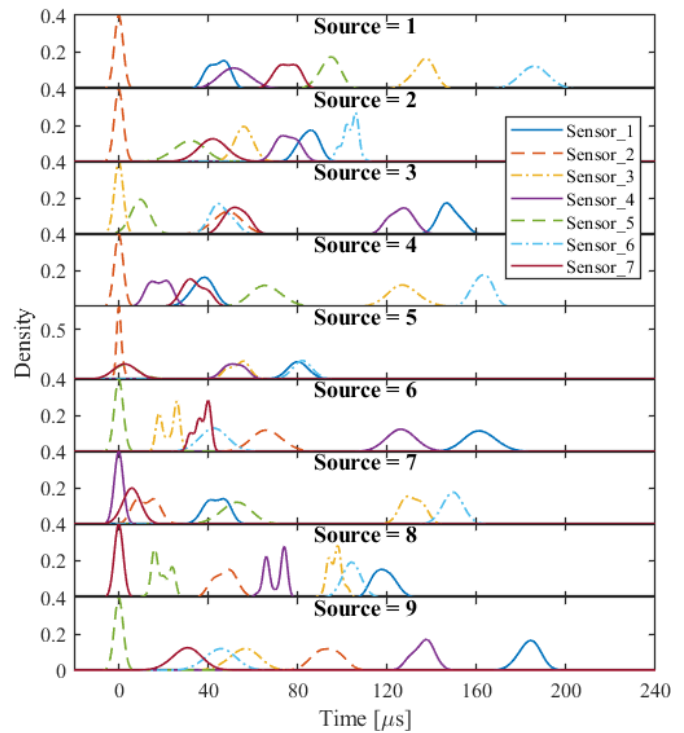


Fig. 14. Distribution of time difference of arrival in sensors for the source location.

C. Source Localization

The AE signal obtained by multiple sensors can be used to estimate the source location based on the difference in the measured features. This feature was considered to be the time difference of arrival of signals between sensors in this study, and machine learning, particularly the naive Bayes classifier, was used to estimate the source location by training the features. The algorithm was trained by applying the time difference of arrival for seven sensors as one input dataset and the location of each source as the output. The ground-truth values for 450 datasets obtained by testing each source 50 times were used for training, and the performance of the algorithm was validated through the arrival times of signals extracted from the tested results of ToA-Net.

The distribution of the input data based on KDE used for training the naive Bayes classifier, that is, the time difference of arrival for each sensor, is shown in Fig. 14. Similar to the conventional zonal location method, the information from the nearest sensor is applied most influentially, and the distribution of arrival times for other sensors allows it to determine the class. Owing to the distributions of input data that are shown to be relatively distinct from others, the naive Bayes classifier is expected to identify the source location well.

Fig. 15 shows the confusion matrix for the source locations estimated using the Naive Bayes classifier. The results show that the algorithm can perform well when estimating the source location using the arrival times of signals extracted from the proposed method, except for the two errors that estimate source location 3 as 6. Considering the distribution of arrival times used for training, sources 3 and 6 have relatively similar distribution characteristics compared with the others. In this

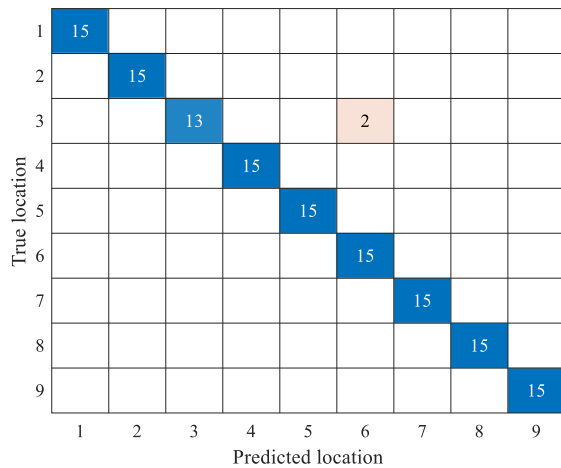


Fig. 15. Results of source location estimation using naive Bayes classifier.

regard, the error is considered to be mostly affected by the order of arrival time in the data from sensors 3 and 5. Allowing that the locations of sources 3 and 6 are adjacent, it can be regarded as an acceptable error.

In this study, an experimental test was performed using a simple specimen, and the sensors were evenly distributed. Accordingly, the adopted probability-based machine learning methodology was able to learn definite features from the input data, and it led to a remarkable performance in the validation test. Although the result would vary depending on the target structure, material properties, or the distribution of sensors, it can derive a probabilistically optimal solution based on pretrained information.

V. CONCLUSION

When a defect occurs inside a structure, identifying its location is significant for the follow-up steps. The AE sensor can detect defect-induced elastic waves, and its location can be estimated using the time information of the signals arriving at multiple sensors. In this study, a hybrid approach was proposed, in which AI technologies are applied step by step for structural damage localization. An LSTM-based ToA-Net was developed to determine the arrival time of AE signals, and a naive Bayes classifier is adopted to estimate the source location through the extracted time information. Since the method allows for the transmission of meaningful physical information in signal processing, it can be regarded to have a variety of applications depending on the intended purpose. The feasibility of the method was validated through experimental tests, and the following conclusions were drawn.

- 1) The LSTM network used in ToA-Net for classifying AE signals and noise showed high performance with an average classification error of approximately 1.5 in time-series data of length 1000. However, the trained networks exhibited discontinuous results in signal classification, which indicates that additional postprocessing is required to determine the arrival times of AE signals accurately.
- 2) AIC can be applied to the data classified by the LSTM network to determine the arrival time of the AE signal.

Although it is difficult to derive the intended output when used alone, AIC works complementary to the LSTM network to achieve its purpose of determining the arrival time of signals.

- 3) The proposed method showed high performance in extracting the arrival time of AE signals with an RMSE of approximately $5.7548 \mu\text{s}$ compared to the ground-truth value. Although this method requires pretraining data according to the applied conditions, the robustness of the method can be increased by preprocessing training data.
- 4) The naive Bayes classifier estimated the source location relatively accurately using the arrival times of the signals extracted through ToA-Net. Although the distribution of the data used in this study was relatively distinct, the method is expected to be able to derive a probabilistically optimal solution based on the trained information, even for other distributions in which the arrival time difference of the signal is not clear.

Although the feasibility of the proposed method has been validated through experiments on a concrete panel, there are still required follow-up studies to enhance the applicability of the method. The studies for improving the method would include source localization for 3-D coordinates rather than predefined zones, testing various locations of sources and sensors, and studying the use of additional information for the signal as well as the arrival time.

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